BU6006 Quantitative Business Methods
Semester 1 Year 2026-2027
Mid-Semester Revision
Time Allowed: 1 hour

Practice Questions

Student Name:	
Student ID:	
Tutor Name:	**Class Number:**

Instructions:

1. Fill in all the required information on the table above.
2. The test consists of two parts:
   - Part 1: 20 MCQs.
   - Part 2: 4 written questions.
3. Students must use the answer sheet for part 1 (MCQ).
4. Round your final answer to 2 decimal places, where required.
5. A scientific non-programmable calculator is allowed.

Learning Outcomes:

1. Apply key statistical concepts, theories, and models applied in quantitative analysis within business and economic contexts.
2. Use descriptive statistics techniques including measures of central tendency, dispersion, and data visualization to present, interpret, and communicate business data effectively.
3. Evaluate business claims using inferential statistics techniques, including probability, distributions, hypothesis testing and confidence intervals, to support evidence-based decision-making in a business context.

Answer Sheet

Part 1: MCQs

Put a check mark (✓) under your chosen choice.

Answers that are not ticked in the table WILL NOT BE MARKED.

Question	A	B	C	D
Example		✓		
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Question	A	B	C	D
Example				✓
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				

Part 1: MCQs
Choose the correct answer.

MCQ1: The weekly sales of a bakery in Bahrain have a mean of BD 600 and a standard deviation of BD 50. According to Chebyshev's theorem, at least what percentage of weeks have sales within 2 standard deviations of the mean?

A. 50%
B. 89%
C. 95%
D. 75%

Use the following information to answer **MCQ2** and **MCQ3.**

A bank branch in Manama recorded the number of customers who visited the branch each day last month.

MCQ2: The type of variable for "number of customers" is:

A. Qualitative (categorical)
B. Quantitative – discrete
C. Both qualitative and quantitative
D. Quantitative – continuous

MCQ3: The level of measurement for "number of customers" is:

A. Nominal
B. Ordinal
C. Ratio
D. Interval

MCQ4: A company received applications from 10 candidates and wants to invite 4 of them for an interview. How many different groups of candidates can be invited?

A. 210
B. 5,040
C. 40
D. 3,628,800

Use the following information to answer **MCQ5** to **MCQ7.**

A sample of 10 gyms in Riffa recorded the number of new memberships sold last month:

34, 41, 28, 52, 45, 38, 30, 47, 36, 49

MCQ5: The Mean ($\bar{x}$) is:

A. 38
B. 41
C. 40
D. 400

MCQ6: The Sample Variance (s^2) is:

A. 8.16
B. 66.67
C. 60.00
D. 600.00

MCQ7: The Sample Standard Deviation (s) is:

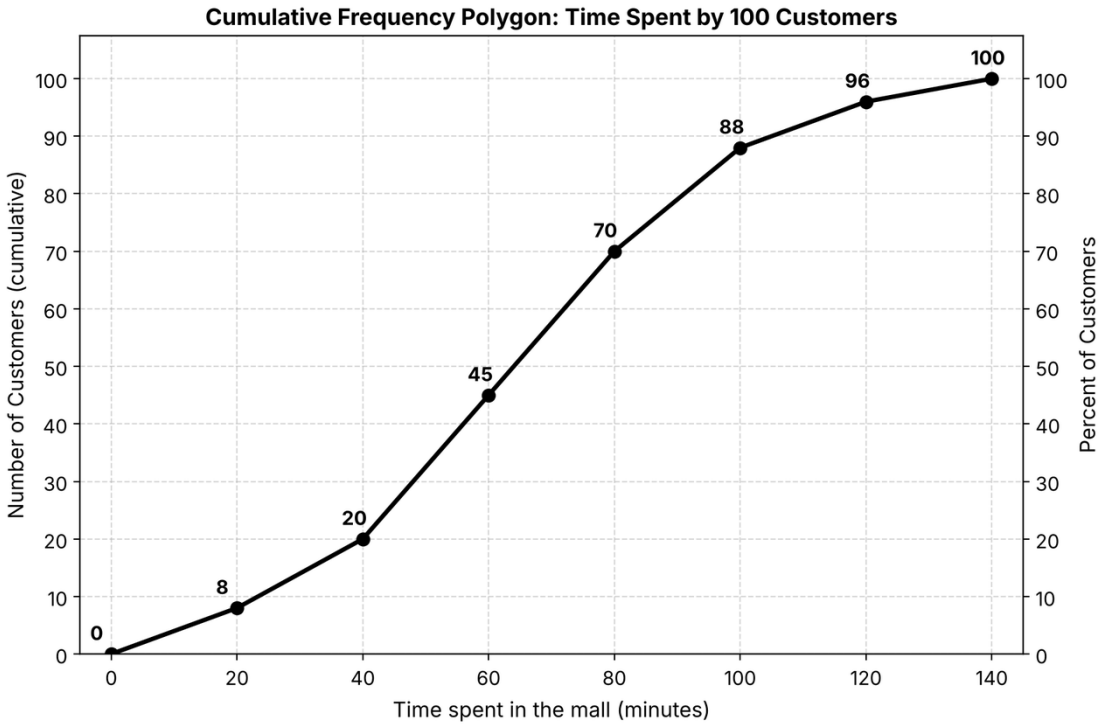
A. 66.67
B. 7.75
C. 40
D. 8.16

MCQ8: The time taken to deliver an online order in Isa Town is uniformly distributed between 10 and 30 minutes.

The probability that an order takes between 15 and 25 minutes to deliver is:

A. 0.50
B. 0.25
C. 0.75
D. 0.83

Use the following information to answer **MCQ9** and **MCQ10.**



Below is the cumulative frequency polygon of the time spent by 100 customers in a shopping mall in Bahrain

MCQ9: The number of customers who spent less than 40 minutes in the mall is:

A. 88
B. 20
C. 8
D. 45

MCQ10: The percentage of customers who spent 100 minutes or more in the mall is:

A. 88%
B. 12%
C. 4%
D. 8%

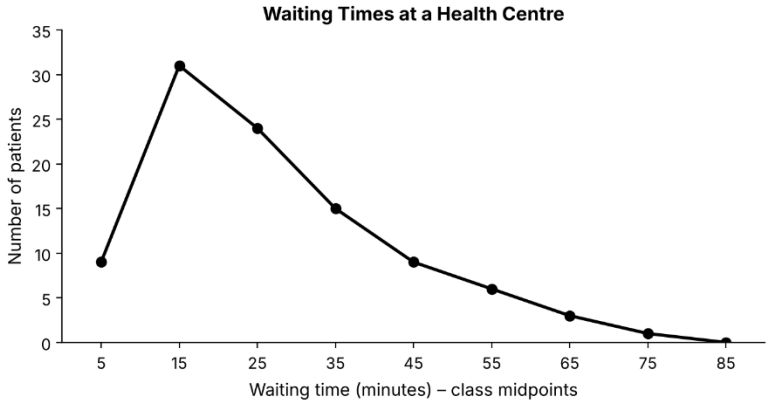
MCQ11: A survey finds that 30% of shoppers in a Bahrain mall use mobile payment. Five shoppers are selected at random. The number of shoppers who use mobile payment follows a binomial distribution. The probability that exactly 2 of them use mobile payment is:

A. 0.36
B. 0.09
C. 0.03
D. 0.31

MCQ12: Four branches of a bank in Bahrain opened the following numbers of new accounts last week: 2, 4, 8 and 16. The geometric mean of the number of new accounts is:

A. 7.50
B. 5.66
C. 1,024
D. 6.00

MCQ13: The graph below shows the waiting times of patients at a health centre in Bahrain.



This graph is a:

A. Histogram, negatively skewed

B. Frequency polygon, positively skewed

C. Frequency polygon, negatively skewed

D. Histogram, positively skewed

The table below represents the frequency distribution of the distance (in km) driven in one day by 60 taxi drivers in Manama.

Distance (km)	Frequency
50 up to 100	6
100 up to 150	14
150 up to 200	22
200 up to 250	12
250 up to 300	6

MCQ14: The number of taxi drivers who drove 200 km or more is:

A. 42

B. 12

C. 22

D. 18

MCQ15: The cumulative relative frequency percentage of taxi drivers who drove less than 250 km is:

A. 54%

B. 90%

C. 10%

D. 20%

MCQ16: A coffee shop in Bahrain recorded the price and the number of cups sold for each cup size yesterday.

Cup size	Price (BD)	Cups sold (w)
Small	0.500	300
Medium	0.800	200
Large	1.200	100
Extra large	1.500	50

Using the number of cups sold as weights, the weighted mean price per cup is:

A. BD 1.00
B. BD 0.78
C. BD 126.25
D. BD 505.00

MCQ17: In a Bahrain mall, 60% of shoppers buy a coffee. If two shoppers are selected at random, what is the probability that neither of them buys a coffee?

A. 0.16
B. 0.36
C. 0.40
D. 1.20

MCQ18: A pharmacy in Muharraq receives an average of 4 online orders per hour. The number of orders per hour follows a Poisson distribution. The probability of receiving exactly 1 order in a given hour is:

A. 0.02
B. 0.15
C. 0.07
D. 0.93

MCQ19: Which measure of central location is most affected by extreme values (outliers)?

A. Mode
B. Mean
C. Median
D. Range

MCQ20: A student club with 8 members wants to elect a president, a vice-president and a secretary. In how many different ways can these three positions be filled?

A. 336

B. 56

C. 24

D. 512

Part 2 : Answer the following questions.

Question 1:

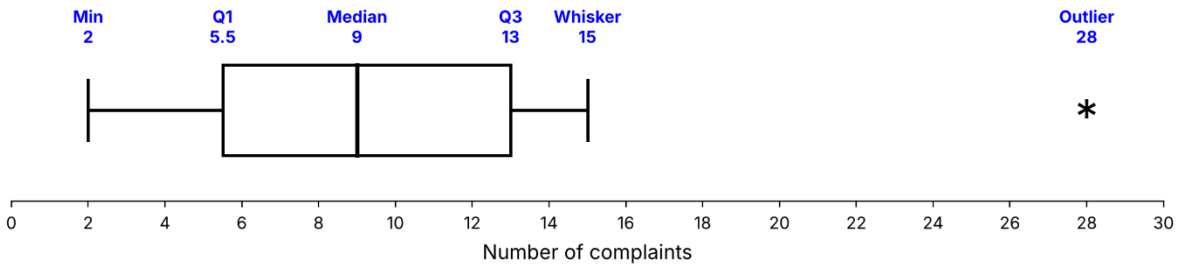
The customer service manager of a bank in Bahrain recorded the number of complaints received last month at 13 branches:

14	2	9	5	11	7	28	15	8	12	6	4	10

a. Find the first quartile (Q1), Median and the third quartile (Q3).
   Q1 = 5.5, Median = 9, Q3 = 13
b. Find the interquartile range (IQR).
   IQR = 7.5
c. Find the 80th percentile P_{80}
   P_{80} = 14.2
d. Find the lower outlier boundary and the upper outlier boundary. Then use them to decide whether there is an outlier in the data. If there is one, identify it.

   Lower boundary = −5.75, Upper boundary = 24.25, so 28 is an outlier.

e. Sketch a box plot for the complaints data on the scale below. Show the minimum, Q1, median, Q3, the end of the right whisker and the outlier.



Question 2:

A manager recorded the time (in minutes) that 20 employees of a company in Bahrain take to travel to work. The times were:

41	22	8	33	16	48	26	13	37	29
11	52	24	36	18	44	27	14	31	21

a. Use the 2^k rule to find the recommended number of classes, then find the class interval, rounding it up to a convenient whole number.
(sorted: 8, 11, 13, 14, 16, 18, 21, 22, 24, 26, 27, 29, 31, 33, 36, 37, 41, 44, 48, 52)
$k = 5$ ($2^4 = 16 < 20 < 32 = 2^5$)
$i \geq (52 - 8) \div 5 = 8.8$, so $i = 10$

b. Using a lower limit of 5 minutes for the first class, construct a frequency distribution.

Travel time (minutes)	Frequency
5 up to 15	4
15 up to 25	5
25 up to 35	5
35 up to 45	4
45 up to 55	2
Total	20

Question 3:

A bank in Bahrain surveyed 200 customers about their preferred banking channel:

Age group	Mobile app	Branch	ATM	Total
Under 30	70	12	18	100
30 and over	40	38	22	100
Total	110	50	40	200

One customer is selected at random. Find the probability that the customer:

a. Prefers the branch.
b. Is under 30 and prefers the ATM.
c. Does not prefer the ATM.
d. Is 30 and over or prefers the mobile app.
e. Prefers the mobile app, given that the customer is under 30.
f. A bank advertisement claims: *"Most of our customers prefer to bank at our branches."*
    i. Convert your answer to part (a) into a percentage and interpret it.
    ii. Is this claim accurate and ethical? Explain.

a. P(Branch) =0.25
b. P(Under 30 and ATM) = 0.09
c. P(not ATM) = 0.80
d. P(30 and over or App) = 0.85
e. P(App | Under 30) = 0.70
f.
i. 0.25 × 100 = 25%, so only 25% of customers prefer to bank at the branches.
ii. Not accurate: only a quarter prefer the branches, and most customers (55%) prefer the mobile app. Not ethical: the advert misrepresents the survey results to promote the branches, which misleads customers.

Question 4:

The time customers spend at a car wash in Bahrain is normally distributed, with a mean of 30 minutes and a standard deviation of 5 minutes.

a. What is the probability that a customer spends between 33 and 40 minutes at the car wash?

   0.2515

b. Using the Empirical Rule, what percentage of customers spend between 20 and 40 minutes at the car wash?

   95%

c. The car wash gives a discount voucher to the 15% of customers who spend the longest time there. What is the minimum time a customer must spend to receive the voucher?

   35.20 minutes

End of Revision Questions

BU6006 Formulas Sheet

Frequency Formulas	Relative Frequency	$\frac{\text{Class Frequency}}{\text{Total Frequency}}$
	Relative Frequency (%)	$\frac{\text{Class Frequency}}{\text{Total Frequency}} \times 100$
Frequency Distributions	Number of Classes	$2^k > n$ or $k > \log_2 n$
	Class Interval	$i \geq \frac{\text{Max} - \text{Min}}{k}$
Measures of Location	Population Mean	$\mu = \frac{\Sigma x}{N}$
	Sample Mean	$\bar{x} = \frac{\Sigma x}{n}$
	Weighted Mean	$\bar{x}_W = \frac{w_1 x_1 + w_2 x_2 + \cdots + w_n x_n}{w_1 + w_2 + \cdots + w_n}$
	Geometric Mean	$GM = \sqrt[n]{(x_1)(x_2)\cdots(x_n)}$
		Scale/Growth Factor $1 \pm r\%$
Measures of Dispersion	Range	Max $-$ Min
	Population Variance	$\sigma^2 = \frac{\Sigma(x-\mu)^2}{N}$
	Population Standard Deviation	$\sigma = \sqrt{\sigma^2}$
	Sample Variance	$s^2 = \frac{\Sigma(x-\bar{x})^2}{n-1}$
	Sample Standard Deviation	$s = \sqrt{s^2}$
Chebyshev's Theorem	$P(\mu - k\sigma \leq x \leq \mu + k\sigma) \geq 1 - \frac{1}{k^2}$ for $k > 1$	
Measures of Position	Location of a Percentile	$L_P = (n+1) \cdot \frac{P}{100}$
	1st Quartile (Q_1) Location	$(n+1)(0.25)$
	Median (Q_2) Location	$(n+1)(0.5)$
	3rd Quartile (Q_3) Location	$(n+1)(0.75)$
	Interpolation for P_n $= (\text{Lower Value}) + (\text{Decimal Part of } L_p)(\text{Upper Value} - \text{Lower Value})$	
	Inter-quartile Range (IQR)	$Q_3 - Q_1$
	Upper Outlier Boundary	$x > Q_3 + 1.5(\text{IQR})$
	Lower Outlier Boundary	$x < Q_1 - 1.5(\text{IQR})$
Empirical Rule	$P(\mu - \sigma \leq x \leq \mu + \sigma) \approx 68\%$	
	$P(\mu - 2\sigma \leq x \leq \mu + 2\sigma) \approx 95\%$	
	$P(\mu - 3\sigma \leq x \leq \mu + 3\sigma) \approx 99.7\%$	

Probability and Counting Rules	Classical Probability	$\frac{\text{Number of favorable outcomes}}{\text{Number of all possible outcomes}}$
	Permutation Formula	$nPr = \frac{n!}{(n-r)!}$
	Combination Formula	$nCr = \frac{n!}{r!\,(n-r)!}$
	Rule of Addition	$P(A \text{ or } B) = P(A) + P(B)$ where A and B are mutually exclusive
		$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
	Complement Rule	$P(A) + P(\text{not } A) = 1$
	Rule of Multiplication	$P(A \text{ and } B) = P(A)P(B)$ where A and B are independent
		$P(A \text{ and } B) = P(A)P(B\|A)$
Continuous Probability Distributions	Uniform Distribution (Density Function)	$f(x) = \frac{1}{b-a}$ if $a \le x \le b$ and 0 elsewhere
	Standard Normal Value (z-value)	$z = \frac{x-\mu}{\sigma}$
		$x = \mu + z\sigma$
Discrete Probability Distributions	Binomial Distribution	$nCx \cdot \pi^x \cdot (1-\pi)^{n-x}$
	Poisson Distribution	$\frac{\mu^x e^{-\mu}}{x!}$